



R3.4 Electron-pair sharing reactions (HL)

Reactivity 3 · What are the mechanisms of chemical change? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Nucleophilic substitution reactions of halogenoalkanes involve a nucleophile (an electron-pair donor, e.g. OH^- , CN^- , NH_3) attacking the carbon bonded to the halogen, which leaves as a halide ion; the mechanism followed depends on the class of halogenoalkane and conditions. $\text{S}_\text{N}2$ (bimolecular) proceeds in a single step, with the nucleophile attacking from the side opposite the leaving halogen (backside attack), causing inversion of configuration, and its rate depends on the concentrations of both the halogenoalkane and the nucleophile — favoured by primary halogenoalkanes (least steric hindrance). $\text{S}_\text{N}1$ (unimolecular) proceeds in two steps via a carbocation intermediate (the halogenoalkane first ionises, then the nucleophile attacks the planar carbocation from either side), and its rate depends only on the concentration of the halogenoalkane — favoured by tertiary halogenoalkanes (more stable carbocation, more steric hindrance to backside attack).

Worked example: follow the reasoning

Outline the $\text{S}_\text{N}2$ mechanism for the reaction of bromoethane, $\text{CH}_3\text{CH}_2\text{Br}$, with hydroxide ions, OH^- , to form ethanol.

Step 1: The OH^- nucleophile (with a lone pair of electrons) approaches the carbon bonded to Br from the side directly opposite the C–Br bond (backside attack), since this carbon is only primary and sterically unhindered.

Step 2: As the new C–O bond begins to form, the C–Br bond simultaneously begins to break in a single concerted step (one transition state, no intermediate); the carbon is briefly five-bonded/trigonal bipyramidal in the transition state.

Step 3: The C–Br bond breaks completely, releasing Br^- , while the new C–OH bond is fully formed; the mechanism proceeds with inversion of configuration at that carbon.

Answer: $\text{CH}_3\text{CH}_2\text{Br} + \text{OH}^- \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{Br}^-$, via a single-step $\text{S}_\text{N}2$ mechanism with backside nucleophilic attack and inversion of configuration.

Quick check · multiple choice

Which halogenoalkane would be expected to react with water predominantly via an $\text{S}_\text{N}1$ mechanism?

- A CH_3Br (a primary halogenoalkane)
- B $\text{CH}_3\text{CH}_2\text{Br}$ (a primary halogenoalkane)
- C $(\text{CH}_3)_3\text{CBr}$ (a tertiary halogenoalkane)
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ (a primary halogenoalkane)

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 2-bromo-2-methylpropane, $(\text{CH}_3)_3\text{CBr}$, reacts with water in a hydrolysis reaction to form an alcohol. [8]
 - (a) Identify the mechanism ($\text{S}_\text{N}1$ or $\text{S}_\text{N}2$) most likely followed by this reaction, giving two reasons based on the structure of the halogenoalkane. [3]
 - (b) Outline the steps of this mechanism, including any intermediate formed. [3]
 - (c) State how the rate equation for this reaction would differ from that of a primary halogenoalkane reacting via the alternative mechanism. [2]

Final answer: _____



Answers and explanation

Question 1 [8]

(a) SN1 mechanism. Reasons: (i) the carbon bonded to Br is tertiary, bonded to three bulky methyl groups, which sterically hinder backside attack needed for SN2; (ii) loss of Br⁻ from this carbon forms a relatively stable tertiary carbocation (stabilised by the electron-donating/inductive effect of three alkyl groups), favouring the two-step ionisation pathway. [3]

(b) Step 1: The C-Br bond breaks heterolytically (spontaneously, without the nucleophile's involvement) to form a planar tertiary carbocation, (CH₃)₃C⁺, and a bromide ion, Br⁻ (this is the slow, rate-determining step). Step 2: A water molecule (nucleophile, using a lone pair on O) attacks the planar carbocation from either face, forming a new C-O bond; loss of H⁺ (to another water molecule) then gives the neutral alcohol product, (CH₃)₃COH. [3]

(c) For this tertiary halogenoalkane (SN1), rate = k[(CH₃)₃CBr] only (first order overall, unimolecular, independent of nucleophile concentration), whereas a primary halogenoalkane reacting via SN2 would have rate = k[halogenoalkane][nucleophile] (second order overall, depending on both concentrations). [2]

Quick check answer

C. Tertiary halogenoalkanes favour SN1 because the tertiary carbocation intermediate is relatively stable and the bulky alkyl groups sterically hinder backside (SN2) attack.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.